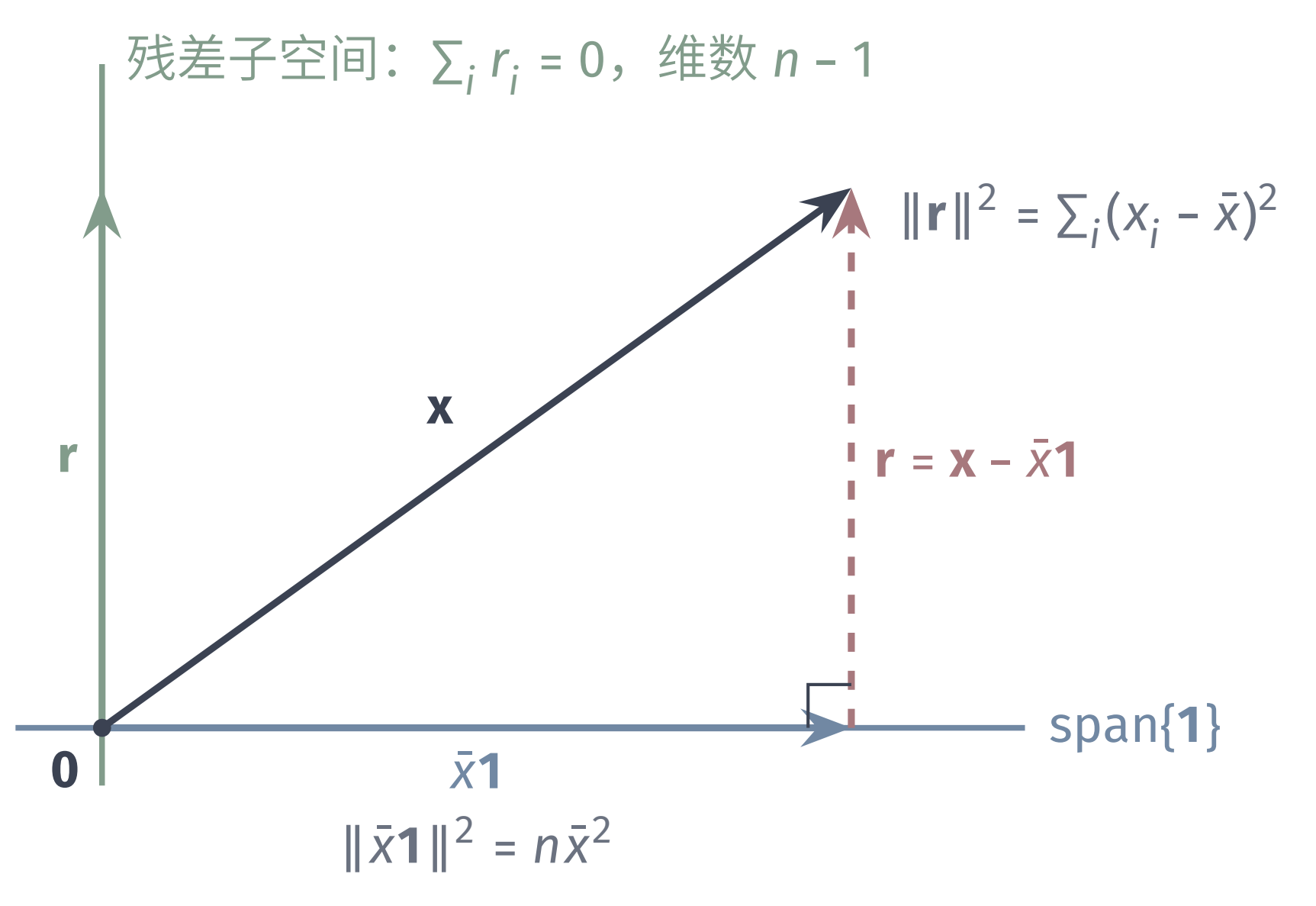
残差子空间：$\sum_i r_i = 0$，维数 $n - 1$
$\|\mathbf{r}\|^2 = \sum_i (x_i - \bar{x})^2$
x
r
$\mathbf{r} = \mathbf{x} - \bar{x}\mathbf{1}$
span{1}
0
$\bar{x}\mathbf{1}$
$\|\bar{x}\mathbf{1}\|^2 = n\bar{x}^2$